

Naveen Balaji

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Profile

Robotics Systems Engineer with hands-on experience in **SLAM, LiDAR-inertial odometry, sensor fusion, GPS-denied navigation, UAV autonomy, mobile manipulation, 3D perception, and robot deployment workflows**. Experienced in building, testing, and debugging autonomy/perception pipelines across simulation and real-world robotic systems, with strong Python/C++ robotics software background.

Core Robotics Skills

SLAM & State Estimation: LiDAR-Inertial Odometry, Visual-Inertial Odometry, Kalman Filters, pose estimation, relocalization, GPS-denied navigation

Sensor Fusion: LiDAR, IMU / inertial sensors, GPS, UWB, BLE, camera-based localization, multi-sensor pose estimation, fail-safe localization

Perception & Mapping: 3D point clouds, PCL, OpenCV, 3D reconstruction, semantic scene completion, SIFT/SURF/ORB, object tracking

Robotics Software: ROS, ROS2, Gazebo, MoveIt, DJI SDK, drone autonomy stacks, PyBullet, OpenAI Gym, Arduino, AWS SageMaker, Linux, Git

Systems & Deployment: robot debugging, hardware-software integration, field validation, testing workflows, logging, experiment tracking, autonomy evaluation

Programming: Python, C++, C, MATLAB, R, PyTorch, $\LaTeX$

Industrial / Mechanical: CAD exposure, mechanical drawing, basic PLC concepts, industrial automation exposure

Education

- Georgia Institute of Technology Atlanta, USA
M.S. in Electrical and Computer Engineering 2024 – 2026
- Georgia Institute of Technology Atlanta, USA
M.S. in Computer Science 2021 – 2023
- Indian Institute of Technology Kanpur Kanpur, India
B.Tech. in Aerospace Engineering 2017 – 2021

Professional Robotics Experience

- Robotics Engineer Atlanta, USA
Joulea LLC May 2023 – Jan. 2024
 - Built and tested a **LiDAR-inertial SLAM and localization pipeline** for autonomous robot operation in large indoor environments, integrating point-cloud mapping, odometry, GPS sensor inputs, and map-based relocalization.
 - Developed **DJI SDK-based drone autonomy workflows** for 3D scanning and building inspection, including flight execution, data capture, point-cloud processing, map generation, and QA of reconstructed environments.
 - Implemented and tested **path-planning and navigation workflows** for drone/robot data collection, integrating GPS, inertial sensing, perception outputs, and mission-level constraints.
 - Integrated perception and localization outputs into **collision-aware navigation workflows**, supporting robot operation in cluttered and partially dynamic building environments.
 - Debugged field issues across GPS, IMU/inertial sensors, mapping, calibration, and autonomy stacks, improving reliability of real-world robot data collection and inspection workflows.
- Team Lead – Robotics Systems Kanpur, India
Aerial Robotics Team, IIT Kanpur Nov. 2017 – Apr. 2019

- Led development of a **vision-based UAV precision landing system** for search-and-rescue scenarios using onboard perception, object tracking, inertial sensing, Pixhawk-based flight control, and autonomous navigation.
- Worked with **Pixhawk-based UAV controllers** for flight-control integration, sensor interfacing, and autonomous mission execution.
- Tuned **DJI SDK-based velocity-control loops** for target tracking and autonomous flight, improving navigation response and control stability.
- Implemented and evaluated **visual-inertial odometry** methods using camera and IMU data for drone navigation in GPS-limited environments.
- Coordinated system integration across perception, path planning, control, GPS/IMU sensing, Pixhawk hardware, and UAV subsystems for competition-ready autonomy.

Research Experience

Robotics Systems Engineer

Guide: Dr. Mangal Kothari

Intelligent Guidance & Control Laboratory, IIT Kanpur

Mar. 2019 – Mar. 2021

- Developed **attitude and position estimators using Kalman filters** for indoor localization and autonomous robot navigation.
- Designed a **GPS-denied fail-safe pose-estimation system** for UAVs by combining inertial and range-based measurements, reducing localization failure cases by 50%.
- Built sensor-fusion algorithms combining **IMU, BLE, and UWB** signals for robust pose estimation, improving localization reliability by 30%.
- Validated localization performance through experimental testing and failure-case analysis across sensing conditions.

Robotics Research Engineer

Guide: Dr. Charlie Kemp

Healthcare Robotics Laboratory, Georgia Tech

Aug. 2021 – Apr. 2023

- Developed simulation and data-generation pipelines for **mobile manipulator autonomy**, supporting training and evaluation of robot learning policies for assistive tasks.
- Built embodied-AI workflows for **grasping, handover, and human-centered robot interaction**, connecting perception outputs with policy learning and task execution.
- Integrated multimodal perception modules for assistive robotics, including visual and sensory inputs for robust human-aware robot behavior.
- Created reproducible experiment workflows for reinforcement-learning evaluations, including simulation setup, policy testing, and qualitative failure analysis.

Robotics / 3D Perception Researcher

Guide: Dr. Lu Gan

Georgia Institute of Technology

Jan. 2024 – Apr. 2024

- Researched **semantic scene completion** for autonomous-robot perception using 3D generative AI models.
- Implemented conditional diffusion baselines with latent-space representations for 3D scene understanding on the KITTI dataset.
- Evaluated 3D perception model performance for robotic scene reasoning, occupancy understanding, and navigation-aware perception.

Machine Learning Research Engineer

Guide: Dr. Anant Madabhushi

Healthcare AI Laboratory, Georgia Tech / Emory

June 2025 – Present

- Developed deep-learning pipelines for medical-image perception, including classification, segmentation, interpretability, and cross-domain validation.
- Built foundation-model feature extraction and survival-modeling workflows for retinal fundus images to estimate time-to-event kidney risk.
- Created heatmap, attribution, and error-analysis workflows to inspect learned image signals and support model debugging.
- Designed reproducible evaluation pipelines for model validation across datasets, improving visibility into failure modes and generalization behavior.

Selected Robotics Projects

Language-Guided Robotic Trajectory Generation

Deep Learning Class Project, Project Report

Jan. – Apr. 2023

- Used CLIP/BERT-style language representations to encode user-defined constraints for robotic trajectory planning.
- Applied diffusion-model methods to learn conditional distributions over feasible trajectory modifications.
- Built simulation workflows for testing planned trajectories on drone platforms.

Learning from Human Feedback for Robot Tasks

Human-Robot Interaction Class Project, Project Report

Jan. – Apr. 2023

- Explored reinforcement learning from human feedback for household robotic tasks using Stretch RE1 simulation environments.
- Compared policies trained with and without human feedback to study user-aligned robot behavior.
- Analyzed how human feedback changes task completion, policy behavior, and failure modes in assistive robotics settings.

Multi-Agent Robotics Planning

Robot Intelligence: Planning, Project Report

Jan. – Apr. 2022

- Built a realistic human-robot simulation environment for coordinated multi-agent task planning.
- Implemented and evaluated PPO, SAC, A2C, Behavior Cloning, and MARWIL for collaborative robot behaviors.

Depth-Camera Gesture Interface

○ *Electronics Club, IIT Kanpur, Project Report*

May – Jul. 2019

- Developed a low-cost human-computer interface using depth cameras and projectors for intuitive robot/computer control.
- Implemented hand tracking and feature-based segmentation using OpenCV, SIFT, SURF, and ORB.

Patent and Publications

Patent: N. Balaji, M. Kothari, and A. Abhishek. System and method for estimation of yaw angle for an aerial vehicle for autonomous navigation, 2019. Indian Provisional Patent

Publication: N. Balaji, M. Kothari, and A. Abhishek. Gps denied localization and magnetometer-free yaw estimation for multi-rotor uavs. In *2020 International Conference on Unmanned Aircraft Systems (ICUAS)*, pages 983–990

Publication: N. Balaji and M. Kothari. Range sensor based Localization and control of mobile robots . SURGE 2019 Poster Presentation, IIT Kanpur [Poster]

Selected Awards

2021: Sri Binay Kumar Sinha Award for best IITK project with industrial applicability and societal impact

2021: General Proficiency Medal for best academic performance at IITK among graduating students

2021: Research Proficiency Medal for best undergraduate project work

2019: Gold Medal, Inter-IIT Tech Meet Aerial Robotics Search & Rescue competition conducted by DRDO

Relevant Coursework

Robotics: Autonomous Navigation, Human-Robot Interaction, Robot Intelligence: Planning, Optimal Control Theory, Unmanned Aerial Systems, Helicopter Theory

CS / AI: Computer Vision, Deep Learning, Machine Learning, Optimization Methods, Data Structures and Algorithms, Graduate Algorithms, Computer Animation

Online: Probability and Statistics, Neural Networks and Deep Learning, Convolutional Neural Networks, Reinforcement Learning Specialization

Teaching and Leadership

Instructor, Human-Robot Interaction Seminar Course

Georgia Tech

- Organized seminar sessions for online master's students and coordinated invited talks from industry and academic robotics/HRI speakers.

Teaching Assistant, Control Systems

Georgia Tech

- Led recitations and hands-on control-system demonstrations, including PID tuning on a toy helicopter platform.

Academic Mentor, Counseling Service

IIT Kanpur

- Tutored students in introductory mechanics and coordinated events/workshops for robotics enthusiasts.